Krishaant Pathmanathan

+1 236 990 4578 | Vancouver, BC | kp243962@gmail.com | linkedin.com/in/krishaant | krishaant.github.io | Writing Portfolio

Education

University of British Columbia September 2022 – December 2027
Bachelor of Arts (B.A.), Economics and Statistics *Vancouver, BC*

- **Relevant Coursework:** Data Science, Statistical Inference, Probabilistic Modeling, Regression Analysis, Causal Inference, Econometrics, Computational Methods, Linear Algebra

Experience

Research Assistant, Software and Data Engineering May 2025 – Present
Vancouver School of Economics *Vancouver, BC*

- Expanded UBC's applied ML curriculum to **5 courses** and **1000+ students** by designing and building **10+ Python and Jupyter notebooks**, covering **image embeddings**, **CNNs**, **network analysis**, and **NLP applications** (sentiment analysis, topic modeling, stance analysis), published on the Praxis Website.
- Built an **LLM**-prediction notebook for **150+ students**, applying **RoBERTa** to score sentiment across **24,000+ financial news and social media texts** and predict 3-month stock returns via linear regression (**R-squared = 0.088**).
- Automated grading for **100+ students**, removing the need for a teaching assistant, by writing a **grading script** that reimplemented **CNN modules** in **raw NumPy** and fixed recurring pipeline failures on the autograding platform.

Machine Learning (NLP) Content Developer Feb 2026 – Apr 2026
Pearson Education *Remote*

- Wrote teaching material on **LLMs** and **NLP** applications (**prompt engineering**, **text classification**, **algorithmic bias**) for an undergraduate sociology textbook, leveraging ML expertise from my UBC research role.
- Built lesson modules, knowledge checks, and interactive exercises aligned to Pearson's curriculum standards.

Data Analytics Intern Jul 2024 – Aug 2024
Entomo Software *Singapore*

- Cleaned and analyzed **5,000+ employee records** in **Python** and **Excel** to track workforce KPIs, then assisted in building a Power BI dashboard for the client.
- Eliminated manual data entry and kept performance dashboards continuously up to date by engineering a scheduled **Python API** integration that pulled **Jira** data and fed automated **Power BI** dashboards, including an **XML-to-JSON** converter for analysis, saving a developer **5+ minutes daily**.

Business Reporter Intern Jan 2023 – Jun 2023
Free Malaysia Today *Kuala Lumpur, Malaysia*

- Reported on **178** corporate and economic stories in **6 months**, publishing **80+** under my own byline (full archive) ranging from market analysis to political economy, including the publication's most-read business article.

Projects

UBC Course Insights | *Python, SQL, DuckDB, Pandas, Streamlit*

- Built a 5-stage ETL pipeline integrating **4 independent sources**, RateMyProfessors, 30 years of UBC grades, UBC's course-schedule API, and professor salaries into a unified **DuckDB** warehouse of **579K+ records** across 9 tables.
- Matched professor identities across sources with inconsistent name formats using a normalized-key system, separating 14,699 unambiguous matches from 439 cases where one name mapped to multiple RateMyProfessors profiles.
- Shipped a full-stack **Streamlit** web app with interactive grade-distribution, rating, and salary visualizations, deployed to a live site. (repo)

Air Quality & Income Inequality: Canadian Metropolitan Areas | *R, OLS Regression*

- Found that worse air quality is linked to higher income inequality across **63 Canadian metro areas** (**adjusted R-squared = 0.78**) by applying **spline interpolation** to 2024 air-quality data and fitting **OLS regressions** in R, validated with **RESET**, **VIF**, and **White's tests**. (repo)
- This paper was presented at LSE's *Knowledge Beyond Boundaries* interdisciplinary research conference. (Jun 2025)

Skills

Languages: Python, R, SQL
Libraries: pandas, NumPy, scikit-learn, Statsmodels, matplotlib, BeautifulSoup
Tools: Docker, Git, GitHub, VS Code, Power BI, Tableau, Excel, Jira, Selenium, LaTex
Concepts: Machine Learning, NLP, CNNs, BERT, Regression Analysis, Feature Engineering, Statistical Modeling